

Task 10: Firewall Configuration & Testing

Firewall Tool Used

- **UFW (Uncomplicated Firewall)**

UFW is a user-friendly firewall tool that works as a front-end for iptables. It helps in managing incoming and outgoing network traffic securely.

1. Learn Firewall Concepts

A **firewall** is a security control that filters network traffic.

It decides:

- What is **allowed**
- What is **blocked**

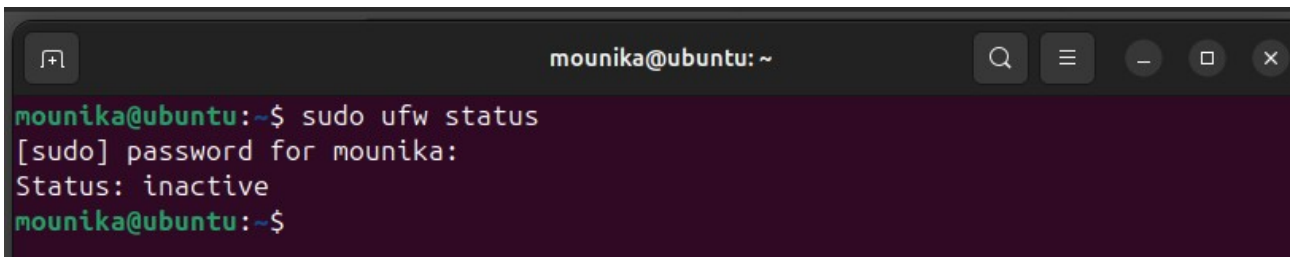
UFW is a firewall tool in Ubuntu that works on top of iptables.

2. Configure Firewall Rules

Check UFW Status

```
sudo ufw status
```

Screenshot : UFW status before enabling

A terminal window with a dark purple background. The title bar shows 'mounika@ubuntu: ~'. The terminal text is as follows:

```
mounika@ubuntu:~$ sudo ufw status
[sudo] password for mounika:
Status: inactive
mounika@ubuntu:~$
```

3. Allow/deny ports

Step 1 — Enable UFW

```
sudo ufw enable
sudo ufw status verbose
```

Screenshot: UFW enabled

```

Status: inactive
mounika@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
mounika@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
mounika@ubuntu:~$
```

Step 2 — Set Default Rules

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
```

Screenshot : Default deny/allow rules

```

mounika@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
mounika@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
mounika@ubuntu:~$
```

All incoming traffic is blocked unless I allow it.

Step 3 — Allow Required Ports

```
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
```

Check:

```
sudo ufw status numbered
```

```
22 – SSH
80 – HTTP
443 – HTTPS
```

Screenshot : Allowed ports

```
mounika@ubuntu:~$ sudo ufw allow 22
Rule added
Rule added (v6)
mounika@ubuntu:~$ sudo ufw allow 80
Rule added
Rule added (v6)
mounika@ubuntu:~$ sudo ufw allow 443
Rule added
Rule added (v6)
mounika@ubuntu:~$
```

Step 4 — Block a Port

`sudo ufw deny 23`

Screenshot : Port 23 blocked

```
mounika@ubuntu:~$ sudo ufw deny 23
Rule added
Rule added (v6)
mounika@ubuntu:~$
```

4. Test connectivity

Ubuntu IP :

```
mounika@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 28:6b:35:31:8f:c6 brd ff:ff:ff:ff:ff:ff
    inet 172.16.97.128/19 brd 172.16.127.255 scope global dynamic noprefixroute wlp0s20f3
        valid_lft 604462sec preferred_lft 604462sec
    inet6 fe80::626d:b652:8226:a2fb/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
mounika@ubuntu:~$
```

```

mounika@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
   Active: active (running) since Fri 2026-01-30 11:49:05 IST; 11s ago
 TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
  Main PID: 9044 (sshd)
    Tasks: 2 (limit: 9001)
  Memory: 2.4M (peak: 2.9M)
     CPU: 34ms
   CGroup: /system.slice/ssh.service
           └─8479 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
             9044 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 30 11:49:04 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell >
Jan 30 11:49:04 ubuntu systemd[1]: ssh.service: Found left-over process 8479 (s>
Jan 30 11:49:04 ubuntu systemd[1]: ssh.service: This usually indicates unclean >
Jan 30 11:49:05 ubuntu sshd[9044]: Server listening on 0.0.0.0 port 22.
Jan 30 11:49:05 ubuntu sshd[9044]: Server listening on :: port 22.
Jan 30 11:49:05 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell s>
lines 1-20/20 (END)

```

Test Connectivity (From Windows VM)

Step 1 — Test Allowed Port

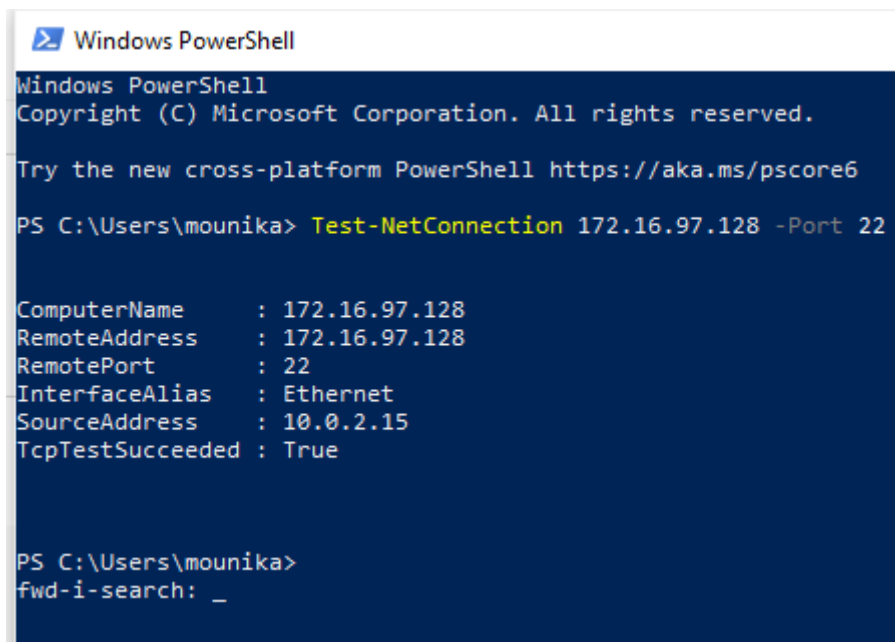
PowerShell:

Test-NetConnection 172.16.97.128 -Port 22

Output:

TcpTestSucceeded : True

Screenshot : Port 22 allowed



```

Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\mounika> Test-NetConnection 172.16.97.128 -Port 22

ComputerName      : 172.16.97.128
RemoteAddress     : 172.16.97.128
RemotePort        : 22
InterfaceAlias    : Ethernet
SourceAddress     : 10.0.2.15
TcpTestSucceeded  : True

PS C:\Users\mounika>
fwd-i-search: _

```

Step 2 — Test Blocked Port

Test-NetConnection 172.16.97.128 -Port 23

Output :

TcpTestSucceeded : False

Screenshot : Port 23 blocked

```
PS C:\Users\mounika> ^C
PS C:\Users\mounika> Test-NetConnection 172.16.97.128 -Port 23
WARNING: TCP connect to (172.16.97.128 : 23) failed

ComputerName           : 172.16.97.128
RemoteAddress          : 172.16.97.128
RemotePort             : 23
InterfaceAlias         : Ethernet
SourceAddress          : 10.0.2.15
PingSucceeded          : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded       : False

PS C:\Users\mounika>
```

5.Observe Firewall Logs (Ubuntu)

sudo ufw logging on
sudo tail -f /var/log/ufw.log

Repeated the test → logs appeared.

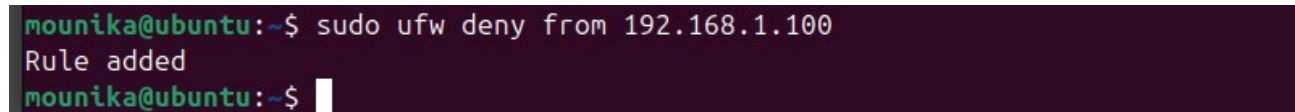
Screenshot : UFW logs

```
mounika@ubuntu:~$ sudo tail -f /var/log/ufw.log
[sudo] password for mounika:
2026-01-30T19:11:21.504447+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=01:00:5e:00:00:fb:6e:0c:ed:a9:64:1e:08:00 SRC=169.254.251.89 DST=224.0.0.251 LEN=32 TOS=0x00 PREC=0x00
TTL=255 ID=46466 PROTO=2
2026-01-30T19:11:55.617337+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=01:00:5e:00:00:fb:6e:0c:ed:a9:64:1e:08:00 SRC=172.16.101.251 DST=224.0.0.251 LEN=32 TOS=0x00 PREC=0x00
TTL=1 ID=19047 PROTO=2
2026-01-30T19:25:51.380384+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=01:00:5e:00:00:fb:6e:0c:ed:a9:64:1e:08:00 SRC=172.16.101.86 DST=224.0.0.251 LEN=32 TOS=0x00 PREC=0x00
TTL=1 ID=51072 PROTO=2
2026-01-30T19:29:34.344440+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=51144 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:34.577398+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=57032 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:34.809450+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=62920 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:35.273440+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=9417 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:36.249345+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=34249 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:38.105431+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=16330 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
2026-01-30T19:29:41.818440+05:30 ubuntu kernel: [UFW BLOCK] IN=wlp0s20f3 OUT= MAC=28:6b:35:31:8f:c6:00:09:0f:09:01:12:08:00 SRC=52.123.128.14 DST=172.16.97.128 LEN=52 TOS=0x00 PREC=0x00
0 TTL=64 ID=45771 DF PROTO=TCP SPT=443 DPT=33666 WINDOW=66 RES=0x00 ACK URG=0
```

6. Block a Malicious IP

```
sudo ufw deny from 192.168.1.100
```

Screenshot : Blocked IP rule



```
mounika@ubuntu:~$ sudo ufw deny from 192.168.1.100
Rule added
mounika@ubuntu:~$
```

7. Document Rules

The firewall rules I configured are documented in the table below:

Rule Type	Rule	Purpose
Default	Deny all incoming	Blocks all unsolicited connections
Default	Allow all outgoing	Allows system to access the internet
Allow	Port 22 (SSH)	Enables secure remote login
Allow	Port 80 (HTTP)	Allows web traffic
Allow	Port 443 (HTTPS)	Allows secure web traffic
Deny	Port 23 (Telnet)	Blocks insecure service
Deny	IP 192.168.1.100	Blocks a suspicious/malicious host

These rules ensure that only trusted services can access the system while all unnecessary and risky connections are blocked.

8. Explain Impact

The impact of these firewall rules is:

- Unauthorized access is blocked
- Only essential services are reachable
- Risk of attacks is reduced
- Malicious IP traffic is stopped
- System network activity is monitored through logs

Overall, the firewall makes the system more secure and controlled by filtering all incoming connections and only allowing what is necessary.